

The Digital Airworthiness Protocol: Establishing a Chain of Custody for Visual Reality

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Abstract

The increasing prevalence of AI-generated images and videos, combined with the ease of manipulating authentic recordings, has fundamentally undermined trust in visual digital media. Images and videos are routinely used as evidence in journalism, legal proceedings, regulatory enforcement, insurance claims, and public discourse, yet there is no universal mechanism to reliably establish whether a given asset represents a real-world capture, a synthetic generation, or an altered derivative.

Current approaches focus largely on detection—attempting to infer whether content is artificial based on visual or statistical cues. This strategy is inherently fragile and does not scale against rapidly improving generative systems. By contrast, high-reliability industries such as aviation do not rely on visual inspection alone; they rely on provenance, traceability, and chain-of-custody to establish trust.

This paper proposes a mandatory, bifurcated provenance framework for digital media, modeled on safety-critical certification practices. The framework consists of two complementary pillars: (1) **Trusted Capture**, in which authorized hardware devices cryptographically sign sensor data at the moment of acquisition to attest to real-world presence; and (2) **Generative Imprint**, in which AI systems embed resilient, cryptographically verifiable identifiers at the point of content generation to attest to synthetic origin.

Together, these mechanisms establish a continuous chain of custody from photon capture or generative synthesis through distribution and consumption. The objective is not to restrict expression or innovation, but to restore verifiable context to visual media in domains where trust, accountability, and evidentiary integrity are essential. *This framework is origin-agnostic: it does not judge the truthfulness of content, only the verifiability of origin and integrity.*

Keywords: Content provenance, AI-generated media, chain of custody, digital evidence, cryptographic signatures, media authenticity, deepfakes, C2PA.

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1 Introduction: The Erosion of Objective Reality

Digital images and videos have long served as a proxy for direct observation. A photograph was assumed to represent a moment in time; a video was assumed to document an event. This assumption is no longer reliable.

Advances in generative artificial intelligence now enable the creation of photorealistic images and videos that depict events, people, and environments that never existed [Ramesh et al., 2022, Rombach et al., 2022]. At the same time, conventional editing tools allow authentic recordings to be altered without leaving perceptible traces. As a result, visual media can no longer be evaluated solely on appearance, even by trained experts.

This shift creates a structural problem. Institutions increasingly rely on visual records for decision-making—ranging from traffic enforcement and insurance assessments to journalism, courts, and regulatory oversight—yet the underlying trust model for those records has not evolved. In many cases, platforms strip metadata, files are recompressed or cropped, and provenance information is lost entirely by the time content reaches an end user.

Most current responses attempt to address this gap through content detection: algorithms that guess whether media is AI-generated based on patterns or artifacts [Rossler et al., 2019]. Detection, however, is an adversarial and asymptotically losing strategy. As generative systems improve, detectable artifacts diminish, and false positives undermine confidence in genuine recordings.

The alternative is provenance. Instead of asking “Does this look real?”, systems must be able to answer “Where did this come from, and how has it changed?”

2 Related Work

This section surveys existing approaches to media authenticity, examining their contributions and limitations. Our proposal builds upon and extends these foundations.

2.1 Content Authenticity Standards

2.1.1 Coalition for Content Provenance and Authenticity (C2PA)

The C2PA consortium, founded by Adobe, Microsoft, Intel, the BBC, and others, has developed an open technical specification for embedding provenance metadata into digital content [C2PA, 2022]. The standard enables cryptographic binding of creator information, capture device data, and edit history to media files. However, C2PA adoption remains *voluntary*, creating a fundamental gap: malicious actors have no incentive to sign fraudulent content, and unsigned content cannot be distinguished from deliberately unmarked fakes.

2.1.2 Content Authenticity Initiative (CAI)

Adobe’s Content Authenticity Initiative [CAI, 2021] extends C2PA into consumer-facing tools and hardware partnerships. Notable implementations include Leica’s M11-P camera with built-in content credentials and integration into Adobe Creative Cloud. While promising, hardware adoption remains limited to premium professional equipment, leaving the vast majority of consumer devices—smartphones, security cameras, dashcams—outside the trust boundary.

2.2 AI Watermarking Approaches

2.2.1 SynthID (Google DeepMind)

SynthID represents a significant advance in robust AI watermarking [Google DeepMind, 2023]. The system embeds imperceptible signals during image generation that survive common transformations in-

cluding compression, cropping, and color adjustment. Critically, SynthID is currently limited to Google’s own generative models (Imagen, Gemini), creating fragmented coverage across the AI ecosystem.

2.2.2 Stable Signature and Invisible Watermarks

Research into steganographic watermarking for diffusion models has produced methods such as Stable Signature [Fernandez et al., 2023], which embeds identifiers directly into the model weights. These approaches show promise for robustness but face ongoing challenges with adversarial removal attacks and lack standardized verification infrastructure.

2.3 Detection-Based Approaches

A substantial body of research focuses on forensic detection of AI-generated content. FaceForensics++ [Rossler et al., 2019] established benchmark datasets for deepfake detection. Subsequent work has explored frequency-domain analysis [Frank et al., 2020], facial physiological signals [Li et al., 2018], and foundation model-based detectors.

However, detection approaches suffer from a fundamental asymmetry: defenders must identify *all* artifacts, while attackers need only eliminate the most detectable ones. As generative models improve, detection accuracy degrades, creating an “arms race” with no stable equilibrium [Groh et al., 2022].

2.4 Regulatory Frameworks

2.4.1 EU AI Act (2024)

The European Union’s AI Act mandates that AI-generated content be clearly labeled [European Parliament, 2024]. Article 50 specifically requires providers of AI systems generating synthetic content to ensure outputs are “marked in a machine-readable format and detectable as artificially generated or manipulated.” While the Act provides regulatory impetus, it defers technical implementation to standards development bodies.

2.4.2 Executive Order on AI Safety (US, 2023)

The United States Executive Order 14110 directs the development of standards for authenticating official government communications and detecting AI-generated content [White House, 2023]. The order emphasizes the need for “robust authentication and labeling” but does not mandate specific technical architectures.

2.5 Gaps in Current Approaches

Table 1 summarizes the key limitations of existing approaches that this paper addresses.

Table 1: Comparison of Existing Approaches

Approach	Mandatory	Hardware Trust	Universal	Resilient
C2PA	No	Partial	Yes	Moderate
CAI	No	Yes	No	Moderate
SynthID	No	N/A	No	High
Detection	N/A	No	Yes	Low
EU AI Act	Yes	No	Regional	TBD
This Paper	Yes	Yes	Yes	High

3 The Philosophy: “Digital Airworthiness”

The fundamental failure of current digital media governance is the assumption that content is innocent until proven guilty—that is, a file is presumed authentic unless a fake is detected. In safety-critical engineering, this logic is inverted. A component is presumed unsafe until its history is verified. This paper proposes adapting the concept of “Airworthiness” to the digital domain.

3.1 The Traceability Imperative

In aviation, a physical component—whether a turbine blade or a simple bolt—is not deemed airworthy based solely on visual inspection. A part may appear flawless, but if it lacks a “Form 1” (EASA Form 1 / FAA 8130-3 Authorized Release Certificate) or equivalent documentation tracing its manufacturing origin and maintenance history, it is legally classified as “unserviceable” [EASA, 2023]. It is rejected not because it is known to be defective, but because its integrity cannot be verified.

Digital media must adopt this standard for high-stakes contexts. An image presented as evidence in a court of law, or as news footage of a conflict, functions as a safety-critical component of the information environment. If such an image lacks a verifiable chain of custody—a cryptographic lineage linking it to a specific sensor or generation event—it should be classified as “unverified data.” It is not necessarily fake, but it is effectively “digital scrap” for evidentiary purposes.

3.2 The Chain of Custody

Trust in a physical supply chain relies on continuity. If a secure container is opened and contents are transferred to an unsealed box, the chain of custody is broken. Similarly, the digital chain of custody must track the asset from the moment of photon capture (sensor) or synthesis (model) through every subsequent state change (editing, compression, transmission).

This principle—that *provenance must be continuous and cryptographically verifiable*—forms the foundation of the Digital Airworthiness Protocol.

3.3 Non-Goals

To preempt potential misinterpretations, this protocol explicitly does **not**:

- **Assess semantic truth:** The system verifies origin and integrity, not whether depicted events are factually accurate.
- **Prevent anonymous speech:** Unsigned content remains publishable; it is simply classified as “unverified” in high-stakes contexts.
- **Block unsanctioned cameras:** Legacy and non-compliant devices continue to function; their output is treated as unsigned.

4 Technical Architecture: The Unified Provenance Model

To operationalize this philosophy, we propose a bifurcated standard that distinguishes between the two fundamental origins of digital media: (1) Physical Reality (Captured) and (2) Computational Synthesis (Generated). Figure 1 illustrates this unified model.

4.1 Pillar A: Trusted Capture (Hardware-Based Truth)

For media claiming to represent physical reality, trust must be rooted in hardware. Software-based verification alone is insufficient, as software environments are malleable.

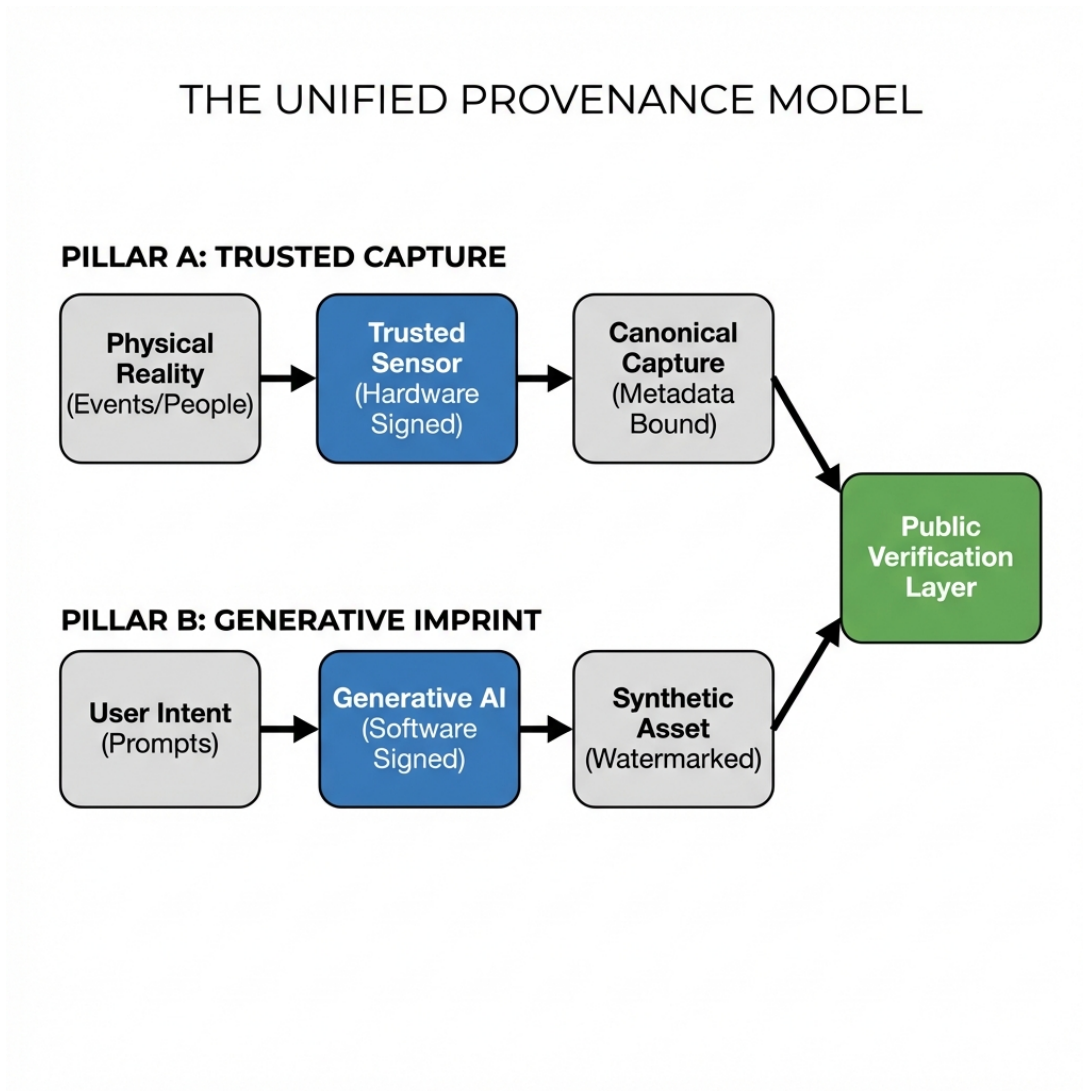


Figure 1: **The Unified Provenance Model.** The framework creates two distinct but convergent paths for digital media: hardware-rooted trust for physical capture (Pillar A) and software-rooted mandates for synthetic generation (Pillar B). Both paths merge at a universal public verification layer.

4.1.1 The Secure Sensor Module

Authorized recording devices (e.g., body cameras, surveillance systems, professional news cameras, and authenticated smartphones) must include a hardware-isolated cryptographic module, similar to a Trusted Platform Module (TPM) [TCG, 2019]. This module—which we term the **Secure Sensor Module (SSM)**—operates within a hardware security boundary that resists tampering and key extraction.

4.1.2 The Signing Process

The signing workflow proceeds as follows:

1. **Acquisition:** Photons impact the sensor array, generating raw data.
2. **Immediate Signing:** Before the data leaves the trusted hardware boundary, it is hashed using SHA-256 and signed using ECDSA with a private key unique to that device.
3. **Metadata Binding:** Critical context—specifically timestamp (from a trusted time source), GPS coordinates (if applicable), and device attestation—is cryptographically bound to the image hash.

4. **Certificate Chain:** The device signature is validated against a manufacturer certificate chain, enabling verification of device authenticity.

4.1.3 The Output: Canonical Capture

The result is a **Canonical Capture**—a cryptographically sealed original that proves *presence*. If a pixel is altered after this signature is applied, the cryptographic hash will fail verification. The system provides:

- **Integrity:** Any modification invalidates the signature.
- **Authenticity:** The signing device is verifiable.
- **Non-repudiation:** The capture event cannot be denied.

4.2 Pillar B: The Generative Imprint (Synthetic Origin)

For media generated by artificial intelligence, trust must be rooted in the generation layer.

4.2.1 Mandatory Encoding

All foundational generative models (large language models with image generation, diffusion models, GANs) above a defined capability threshold must operate under a mandate to embed a “Genesis ID” during the inference process. This identifier includes:

- Model identifier and version
- Generation timestamp
- Generating organization
- Unique output hash

4.2.2 Resilient Watermarking

The imprint must utilize advanced steganographic techniques that distribute the signal across the entire file. Building on research in robust watermarking [[Fernandez et al., 2023](#), [Google DeepMind, 2023](#)], the standard specifies:

- **Frequency Domain Embedding:** Signals encoded in DCT/DFT coefficients survive recompression.
- **Spatial Redundancy:** Critical bits replicated across image regions survive cropping.
- **Perceptual Hashing:** Watermark remains detectable even after analog reconversion (e.g., photographing a screen).

4.2.3 Universal Decodability

The standard must be readable by open-source libraries integrated into standard web browsers and operating systems, avoiding proprietary vendor lock-in. Reference implementations should be published under permissive licenses.

5 The Verification Layer: User Accessibility

A provenance standard is useless if it is invisible to the end user. The verification layer acts as the interface between the cryptographic backbone and the human observer.

5.1 The “Traffic Light” System

We propose a standardized UI/UX indicator for social media platforms, media players, and web browsers:

- **Green (Verified Capture):** The media bears a valid, unbroken hardware signature from a known device manufacturer. (Context: “This is a verified photograph.”)
- **Yellow (Verified Synthetic):** The media bears a valid software signature from a known generative model. (Context: “This content was AI-generated.”)
- **Red / Gray (Unverified):** No signature is present, or the signature is broken. (Context: “Origin cannot be verified.”)

This visual language is intentionally simple, providing immediate context without requiring technical expertise.

5.2 Detailed Provenance Inspection

For users requiring deeper verification—journalists, legal professionals, investigators—the system provides full provenance inspection:

- Complete certificate chain for hardware-signed content
- Model and organization details for AI-generated content
- Full edit history where available
- Geographic and temporal attestations

5.3 Resilience and Recovery

In scenarios where file metadata is stripped (e.g., by legacy platforms that do not preserve C2PA manifests), the system relies on perceptual hashing. A user inspecting a stripped image can query a public ledger using the image’s perceptual hash. If the original signed version exists in the ledger, the system can retrieve the original context and provenance data, closing the loop even when the file itself has been degraded.

6 Threat Model and Security Considerations

Any provenance system must anticipate adversarial attacks. This section addresses primary threat vectors.

6.1 Hardware Key Compromise

If a device’s private signing key is extracted, an attacker could forge “trusted” captures. Mitigations include:

- Hardware Security Modules (HSM) with tamper-evident enclosures
- Remote attestation and periodic key rotation
- Revocation infrastructure for compromised devices
- Rate limiting and anomaly detection on certificate authorities

6.2 Watermark Removal Attacks

Adversaries may attempt to strip or overwrite generative watermarks. Defense-in-depth strategies include:

- Multi-layer watermarking (visible metadata + invisible steganography)
- Watermark designs robust to common transformations
- Legal penalties for deliberate watermark removal (as established in EU AI Act)

6.3 Laundering Through Capture

An attacker might display AI-generated content on a screen and photograph it with a trusted device, creating a “real” capture of synthetic content. Mitigations include:

- Screen detection algorithms in the signing pipeline
- Moiré pattern and refresh rate analysis
- Metadata cross-referencing (claimed location vs. device history)

6.4 Coercion and Misuse

Trusted capture could enable surveillance misuse. The architecture must:

- Separate authenticity from identity (prove device validity without revealing photographer)
- Support selective disclosure (verify time without revealing location)
- Enable user-controlled redaction that preserves integrity proofs for remaining content

7 Implementation Strategy and Policy Recommendations

7.1 The Mandate Argument

Voluntary standards face an insurmountable “Prisoner’s Dilemma”: bad actors will never voluntarily sign fraudulent content, while good actors bear adoption costs. This dynamic ensures that unsigned content remains ubiquitous, undermining the entire system.

Therefore, regulatory mandates are required for the *tools of creation*:

7.1.1 Hardware Mandate

- **Phase 1 (2026-2027):** Professional equipment (broadcast cameras, body cameras, forensic devices) must include SSM.
- **Phase 2 (2028-2029):** Consumer smartphones and tablets must include SSM for camera applications.
- **Phase 3 (2030+):** All imaging devices above a minimum resolution threshold must comply.

7.1.2 Software Mandate

- Commercial distribution of generative AI models lacking mandatory watermarking should be prohibited in regulated markets.
- Open-source models should be required to include watermarking in reference implementations.
- API providers must ensure watermarking is applied regardless of downstream usage.

7.2 Privacy Considerations

Trusted Capture raises legitimate surveillance concerns. The standard must architecturally separate *authenticity* from *identity*:

- A camera proves it is a “valid Sony A7R V” without revealing owner registration.
- Timestamps are cryptographically verifiable without requiring location disclosure.
- Privacy-sensitive data (precise GPS) can be hash-locked—verifiable only if the creator chooses to reveal the key.
- Content can be published with redacted metadata while retaining image integrity signatures.

7.3 Governance and Standards Bodies

Implementation requires coordination across:

- **Technical Standards:** ISO, IEEE, and C2PA working groups
- **Hardware Certification:** Device manufacturer consortia
- **Regulatory Enforcement:** EU, US, and other national regulators
- **Platform Integration:** Social media and content distribution networks

8 Conclusion

The erosion of trust in digital media is not a future risk; it is a present condition. Addressing it does not require banning generative technologies or centralizing control, but engineering a transparent and verifiable provenance layer for visual reality.

By unifying AI-generated media provenance with trusted real-world capture signatures, society can regain the ability to distinguish creation, alteration, and authentic recording. The aviation industry demonstrates that safety-critical systems can operate reliably at global scale through rigorous chain-of-custody requirements. The same principles apply to the information environment.

We regulated the physical sky to ensure safety; we must now regulate the digital cloud to ensure truth.

8.1 Call to Action

1. **Standards Bodies:** Extend C2PA to include mandatory hardware attestation requirements.
2. **Regulators:** Implement phased mandates for both capture devices and generative systems.
3. **Hardware Manufacturers:** Integrate Secure Sensor Modules into imaging product roadmaps.
4. **AI Developers:** Adopt robust watermarking as a non-negotiable component of model deployment.
5. **Platforms:** Implement universal provenance display (“traffic light” indicators) in content interfaces.

The technology exists. The standards are emerging. What remains is the collective will to make provenance mandatory—to establish, at last, a chain of custody for visual reality.

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